

# Leveraging YOLO models for Enhanced Object Detection and Recognition in E-Commerce

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**Abstract** — Visual object detection and recognition is playing a vital role in many majorities of the e-commerce platforms now-a-days. Especially after the COVID-19 pandemic era, most of the developing nations are moving towards digitalization, and internet usage is rapidly increasing day by day. The growing significance of purchases through e-commerce platforms has led to increased popularity, where the adoption of various Artificial Intelligence techniques, such as real-time visual product detection and recognition, remains a key challenge in addressing customer needs. This paper focuses on visual object detection and classification using different deep learning models like YOLOv8, YOLOv9, and YOLO-NAS, utilizing Google's Open Image Dataset with five distinct classes and a custom-curated dataset containing four classes from popular Indian e-commerce platforms. These models have been thoroughly tested using various performance metrics, including Recall, Precision, mean average precision, and F1-Score, with the YOLOv9m model achieving a 97.8% mAP on the custom dataset. Further, our fine-tuned YOLOv9m model was cross-validated using various unseen e-commerce images for detection and classification, which provided promising results.

**Keywords**—Object Detection, Visual Recommendation, YOLOv9, Image Classification, Product Identification.

## I. INTRODUCTION

The role of Artificial Intelligence is increasing rapidly and is playing a vital role in the majority of applications. e-Commerce is one such application where the use of Artificial Intelligence and Machine Learning advancements, strengthen the personalized recommendations, customer service, inventory management and many more. During and after the COVID-19 pandemic, the purchase of commodities on e-commerce platforms has been increasing and is improving due to customers' rise. Most of the e-commerce platforms works on the basis of the text based product recommendations. It is mainly relying on the search queries, and product descriptions. This results in fetches the incomplete or inappropriate recommendations. It is important to understand the visual objects in the products based on its various features. Few e-commerce platforms are utilizing the visual product recommendations such as Amazon, Flipkart, Mytra, Meesho etc for recommending the products to the end users. One of the great challenges in visual based object detection and recognition in the e-commerce platforms is accuracy in detection and classification. This involves in detecting various multi-visual features such as color, shape, texture, pattern and design. Furthermore, these e-commerce platforms are listing various product images taking from multiple vendors and images

vary with different lighting contrast, backgrounds, and resolutions. When models are trained with standard datasets and compared with real world product images it may leads to misclassifications of the products. Existing object detection models [1] [2] [3] suchas YoloV3, YoloV4, Faster-RCNN, SSD etc struggle with balancing the precision, accuracy and speed when trained on large e-commerce product datasets.

In the process of personalized recommendation, object detection helps identify and locate the objects within the images or videos. Many e-commerce platforms offers the personalized visual product recommendations especially for the fashion products suchas shirt, jeans, cap, belt, watch etc. To address these challenges, this research proposes an multi-visual object detection using real-world e-commerce product images which are custom curated from various Indian e-commerce platforms. This aims to develop the high accurate, computationally better efficient object detection model using various Yolo based models such as YoloV8, Yolov9 and Yolo-NAS for e-commerce products.

Detecting an object within an image by identifying the object based on object classes involves information like the location of an object, scale bounding box, or segmentation task. Generative or Discriminative models are considered as the traditional methods for object detection [4]. Generative models rely on probability, variability, and appearance modeling to represent image appearance, while discriminative models focus on identifying objects within images by training classifiers. Object detection plays a critical role across various domains, with benchmarks set by the COCO object detection challenges. Techniques like Mask R-CNN have further expanded detection capabilities in augmented reality and video conferencing. In [5], the YOLO approach was framed as a regression problem for object detection, utilizing full images to identify bounding boxes and class probabilities. This approach notably enhanced the speed and efficiency of the pipeline. While YOLO typically processes images at 45 frames per second, the authors employed a smaller variant, Fast YOLO, which processes at 155 frames per second while still maintaining high accuracy. In [6], an application was created to register users through image captures, utilizing four open-source facial detection algorithms with the Wider Face and MegaFace datasets. The authors assessed both the Single Stage Headless Face Detector and R-CNN, discovering that while SSH offered superior precision-recall performance, it operated more slowly than R-

CNN. They also tried it on InsightFace and Dlib, where InsightFace gives a five-time better precision-recall performance. [7] Developed a data set titled Open Images Dataset V4 with helps for the researchers working on Computer Vision who requires large scale image datasets for training and evaluating, for the image classification, detection of objects and their relationship. The dataset consists of millions of labelled images and the thousands of object classes and the relationship between the objects [8].

Object recognition by the computers and how the new methods of detecting the objects increased with the help of technology like deep learning. They discussed different ways for detecting the objects and training the systems [9] [10]. They also discussed different methods used to spot the single and multiple objects. In [11] Explored that various different collaborative filtering techniques are developed so as to enhance the recommendation systems by reducing search costs when compared with the traditional approaches. The authors introduced a traditional collaborative filtering approach, utilizing two novel similarity models namely Relevant Jaccard Mean Square Distance and Relevant Jaccard Similarity. They used MovieLens dataset as the benchmark in their recommender system domain by leveraging the rating vectors of users to improve the accuracy with less computation time.

Few authors analyzed numerous methods for object detection and segmentation of apparel for the fashion images. They proposed two models for the namely convolutional neural network and Mask R-CNN models on the Fashion-MNIST dataset and found 93% accuracy with the CNN model [12]. In [13], the authors enhanced Contrastive Predictive Coding through unsupervised learning by improving linear classification accuracy on the ImageNet dataset. They achieved this using 2-5X fewer labels compared to classifiers trained directly on image pixels. This method was also applied to the PASCAL VOC dataset with supervised pre-trained ImageNet classifiers.

In [14], the authors developed a system to automatically detect individuals in thermal images using CNN models originally designed for RGB image identification. They evaluated several object detection methods, including SSD, Faster R-CNN, Cascade YOLOv3, and R-CNN, also determined that YOLOv3 was significantly faster in identifying objects according to their experiments.

In [15] Propounded PeleeNet a smart network that runs on phone which works faster and takes less amount of space to recognize objects in the photos or by speech understanding. They performed testing with various networks and found that their system recognizes the pictures faster. In [16], a new real-time object detection method named YOLOv4-tiny was introduced, which builds on YOLOv4. This approach incorporates two ResBlock-D modules from the ResNet-D network to lower computational complexity and includes an auxiliary residual network block to enhance object information extraction and reduce detection errors. Experimental results showed that YOLOv4-tiny achieved faster object detection compared to YOLOv3-tiny. In [17], the authors explored object detection with a focus on optimizing neural network architecture. They introduced several key enhancements, including a weighted Bi-directional Feature

Pyramid Network (BiFPN) for optimized multi-scale feature integration, and a compound scaling technique that simultaneously adjusts the resolution, depth, and width of the backbone, feature network, and prediction layers to improve performance. These optimizations resulted in the EfficientDet family of object detectors, which offered improved efficiency. In [18] provides a brief detail of detecting an object through the deep learning approaches which involves generic object detection like detection frameworks, representation of an object feature, and generation of object proposal, context modelling, different training strategies and evaluation metrics. [19] Analyzed different methods used to detect the objects in early stages of the research and how it is evolved by recognizing using the deep learning techniques. In this the authors discussed the keen points on recognizing the objects, training the models, basic parts of the recommendation system and different techniques to make the system work faster and novel ways for recognizing the objects.

In [20], a model was proposed that detects objects within an image by generating bounding boxes and predicting a confidence score for each box. Ultimately, the box with the highest score is selected as the most accurate detection. If a newly identified box proves more efficient than a previously recorded one, the algorithm automatically replaces the old box with the new one. [21] Improved Grounding DINO was presented to enhance clothing object detection using the Garment40K dataset, which includes textual descriptions and category labels. By adjusting constraints in the adapter module parameters of the existing model, the authors improved performance while lowering computational costs. [22] A product recommendation system for e-commerce was proposed, integrating Case-Based Reasoning with K-means clustering. CBR provides recommendations based on users' transaction histories, while K-Means Clustering identifies and suggests products with similar preferences. The system's effectiveness was validated by comparing its recommendations to user choices, demonstrating improved performance. In [23] designed computer vision systems for object detection, introducing two distinct types of detectors: single-stage and two-stage. The single-stage detectors process the image in a single pass, whereas the two-stage detectors employ a region-based approach, focusing on specific areas of interest. The paper also discussed the strengths of different object detection methods.

This paper evaluates the various Object detection models using Yolov8, Yolov9 and Yolo-NAS architectures for object detection, recognition and classifications for e-commerce product images.

- The main objective is to design the multi-visual object detections for e-commerce products using OIDV4 dataset and custom curated dataset generated from Indian e-commerce platforms.
- The model primarily implements the fined tuned custom Yolov9-M model with the CSPDarkNet53 backbone architecture utilizing the Programmable Gradient Information (PGI), RepConvolution Blocks, Spatial Pyramid Pooling for better and efficient object localization and extraction. Further both the datasets i.e, OIDV4 and custom datasets are evaluated

on the YOLOv8, YOLOv9 and YOLO-NAS variants for comparison.

- The proposed model is evaluated using various metrics such as mean average precision (mAP), Precision, recall for individual classes. In our approach a combined loss functions such as localization loss, confidence loss, multi-visual classification loss and auxiliary loss is computed for enabling the model to perform realtime object detection for fast and reliable e-commerce product recognition and classification.

Further in the subsequent sections, Section 2 introduces the proposed methodology, Section 3 shows the Experimentation and Results, Section 4 includes discussion on results and Section 5 draws the final conclusion.

## II. RESEARCH METHOD

In this paper, to implement multi-visual object detection, recognition, and classification, we have implemented various latest-generation YOLO models such as YOLOv9 and YOLOv8 to analyze the performance and consistency in detecting the objects using OVIDV4 dataset and Custom curated image product image datasets from various Indian e-commerce portals. In the Figure 1, clearly described the process of multi-visual object detection process and further we have mainly adopted the YOLOv9m model with the CSPDarknet53 as backbone architecture. The YOLOv9 [24] architecture begins with a foundational convolutional block featuring a 2D convolutional layer, 2D batch normalization, and the SiLU activation function, setting the stage for subsequent blocks in the network. All these together fuse to form the convolution block of the YOLOV9 architecture. The YOLOV9 architecture has an auto pad code, which is used to determine the padding value when using a convolution block. The main components in YOLOV9 such as Backbone, Neck, Head, Auxiliary branch, Programmable gradient information, generalized efficient layer aggregation network. The following Figure 2, describes the inner working process of YOLOV9m using e-commerce product image dataset. The RepConvolutional block [25] consists of two convolutional layers, element-wise addition, and a SiLU activation function.

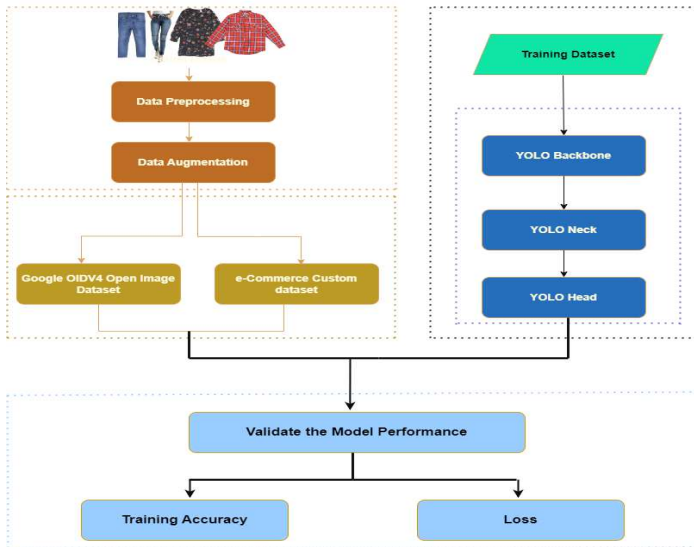


Fig 1. Proposed Process Model for Multi-Visual Object Detection using YOLOv9M

The input undergoes initial processing through a pair of convolutional layers: the first layer employs a  $3 \times 3$  kernel with a stride of 1 and padding of 1, while the second layer uses a  $1 \times 1$  kernel with a stride of 1 and no padding. The outputs from these two layers are then combined through element-wise addition, followed by the application of the SiLU activation function to the resulting sum.

In contrast to YOLOV8, the YOLOV9 has mainly information bottleneck principle focusing on the information loss in neural network, Reversible function ensures no loss during data transformation, Programmable Gradient information (PGI) RepNCSPPELAN4 block which ensures the consistent gradient loss calculation. Later, Generalized Efficient Layer Aggregation Network (GELAN) [26] combines two neural architectures CSPnet and Efficient Layer Aggregation Network (ELAN) for enhancing and understanding the e-commerce product images for efficiently. GELAN generalizes ELAN capabilities. This block includes a conventional block, which splits the resulting feature maps. One set of feature maps is directed to the RepNCSP [27] block convolutional block, while the other set goes directly to the concat block.

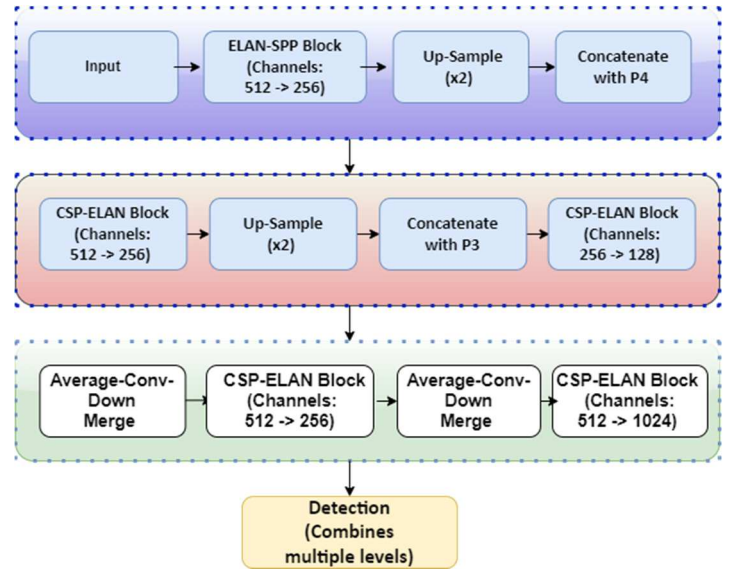


Fig 2. YOLOv9M Internal Layer Architecture

The Adown block is used for down sampling operations or reducing the resolution of the feature map. In this block, the processed input from the preceding block is down sampled using two-dimensional average pooling, decreasing the resolution of the feature map while preserving key features. The results will be divided into two parts. One part will go to the convolution block while the other will go to the max pooling followed by another convolutional block. Output results from the two paths will be combined again using the concatenation operation.

The next block is the Spatial Pyramid Pooling (SPPELAN) [28] block and It is a modification of regular SPP block for better data processing during detection and classification task. The primary function of SPPELAN is to create a fixed feature representation of objects of varying sizes in an image without the need to resize the image or lose spatial information. In this block, there is a convolutional block followed by 3 SP blocks. These SP blocks contains the MaxPool layer, every resulting feature map is concatenated before and is sent to the convolutional block at the end of SPPELAN block. The following algorithmic steps used to compute the multi-visual object and classification using e-commerce product images.

## Algorithm to compute Multi-Visual Object Detection and Classification

*Input: E-commerce Product Image*

*Output: Multi Object Detection and classification*

Pre-Processing:

1. Load the image dataset
2. Image Augmentation

Feature Extract and Data Processing:

1. Extracting features using CSPDarkNet53 backbone architecture
2. Feature Aggregation under Neck block
3. Process for Predictions using Detection Head block
4. Bounding box predictions
5. Computing the loss such as localization loss, confidence loss, and Classification loss
6. Apply Non-Maximum suppression
7. Training and Optimization process
8. Final outcome with class lables with bounding box.

We have calculated the loss function for the proposed model using the following equations. Equation (1), describes the regression loss for bounding box, equation (2) describes the objectness loss using binary cross entropy, equation (3) describes class prediction loss for multi-visual classification task and finally computes the additional loss component for auxiliary loss for efficient detection performance using equation (4).

$$L_{box} = \lambda_{coord} \times \sum (C_{IoU}(x_{pred} - x_{true}, y_{pred} - y_{true}, w_{pred} - w_{true}, h_{pred} - h_{true})) \quad (1)$$

$$L_{obj} = \sum (BCE(obj_{pred}, obj_{true})) \quad (2)$$

$$L_{cls} = \sum (CrossEntropy(class_{pred}, class_{true})) \quad (3)$$

$$L_{aux} = \sum (FocalLoss(class_{pred}, class_{true})) \quad (4)$$

### III. EXPERIMENTATION AND RESULTS

For object detection and classification, the YOLOv9 experimentation was explored with the significant focus on gathering two different datasets consisting of fashion apparel products. In the initial approach, the data set was obtained from Google Open Image Dataset (OIDV4), consisting of 1100 images of five different object classes, which consist of trousers, jeans, shirts, belts, and hats. OIDV4 is a comprehensive open-image dataset featuring over 1.9 million of images and 15.4 million bounding boxes over an extensive range of object categories. Additionally, this dataset, a customized dataset was manually created by crawling images from various e-commerce platforms, including Myntra, Flipkart, Amazon, and Meesho. A total of 1200 images were manually curated for five different object classes like t-shirts, trousers, jeans, hats and belts.

The custom data set offers accurated selection from E-Commerce platforms potentially providing cleaner and more relevant data for training and evaluation purpose. The experimentation is carried out using Intel Core I5 processor with 16gigabytes of RAM and RTX 2050 graphics card with four gigabytes of RAM and utilized the python customized environment using PyCharm framework. The final experimentation results are generated using the YOLOv9m architecture model and evaluated

using various performance metrics such as Precision, Recall, and Generalized Intersection over union (GIoU) and Mean average precision for object detections. The following mathematical equations (5), (6) and (7) are used to determine the various class probabilities and model analysis.

$$Precision = TP / (TP + FP) \quad (5)$$

$$Recall = TP / (TP + FN) \quad (6)$$

$$mAP = \frac{1}{N} \sum_{i=1}^N AP_i \quad (7)$$

Table 1 presents the various class probabilities computed using the YOLOv9m model for e-commerce object detections on the custom dataset and Table 2 results represents the OIDV4 dataset and custom-curated datasets from various Indian e-commerce platforms.

Table 1. YOLOv9m model performance on Custom dataset

| Class           | Precision | Recall | mAP50 | mAP50-95 |
|-----------------|-----------|--------|-------|----------|
| Kids-boys-jeans | 0.947     | 0.977  | 0.976 | 0.733    |
| Kids-boy-shirt  | 0.99      | 0.991  | 0.994 | 0.801    |
| Kids-girl-dress | 0.943     | 0.975  | 0.97  | 0.766    |
| Women-jean      | 0.985     | 0.877  | 0.97  | 0.645    |

Table 2. YOLOv9m model performance on OIDV4 dataset

| Class    | Precision | Recall | mAP50 | mAP50-95 |
|----------|-----------|--------|-------|----------|
| Shirt    | 0.39      | 0.526  | 0.455 | 0.314    |
| Trousers | 0.281     | 0.404  | 0.265 | 0.167    |
| Jeans    | 0.548     | 0.344  | 0.354 | 0.249    |
| Belt     | 0.102     | 0.615  | 0.482 | 0.279    |
| Hat      | 0.657     | 0.556  | 0.614 | 0.426    |

Furthermore, we computed and compared the performance of various benchmark algorithms using YOLOv8 and YOLOv9 on both the OIDV4 and the custom e-commerce datasets. Table 3 presents the overall performance metrics alongside the Giga Floating Point Operations per Second (GFLOPs) measure for a clearer understanding of the computational efficiency. This allows for evaluating the significance of computational calculations and GPU utilization in the context of these algorithms.

Table 3. Quantitative performance results using various Yolo Models

| Performance measures using E-commerce Custom Dataset |           |        |       |          |        |
|------------------------------------------------------|-----------|--------|-------|----------|--------|
| Model                                                | Precision | Recall | mAP50 | mAP50-95 | GFLOPs |
| YoloV9M                                              | 0.966     | 0.955  | 0.978 | 0.736    | 76.5   |
| YoloV9S                                              | 0.901     | 0.923  | 0.934 | 0.698    | 75.7   |
| YoloV8M                                              | 0.911     | 0.90   | 0.912 | 0.65.4   | 69.5   |
| Performance measures using OIDV4 apparel Dataset     |           |        |       |          |        |
| YoloV9M                                              | 0.441     | 0.432  | 0.433 | 0.292    | 8.1    |
| YoloV9S                                              | 0.396     | 0.489  | 0.434 | 0.287    | 8.1    |
| YoloV8M                                              | 0.366     | 0.389  | 0.378 | 0.243    | 7.3    |
| Yolo-NAS-M                                           | 0.40      | 0.429  | 0.43  | 0.31     | 8.1    |



The following Figure 3, shows the various loss functions, metric results such as Precision, mean average precision at various conditions and recall function using YoloV9M for our custom curated dataset.

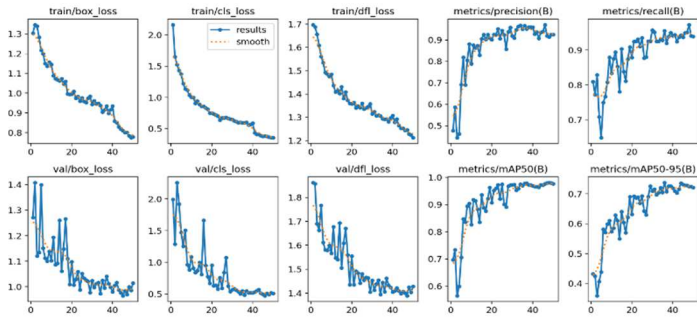


Figure 3. YoloV9m Model visualization for custom e-commerce image dataset.

Finally, we are presented the various results after finetuning the YoloV9m model on the custom image dataset and sample results are shown in the Figure 4.



Figure 4. Sample detection and classification from input images

#### IV. DISCUSSIONS

The YOLOv9m model significantly performed on the custom curated dataset, showing high precision and recall rate are 0.99 for categories like kids-boy-shirts. Whereas the model is shown the significant precision, recall and mAP on the remaining category of products such as kids-boys-jeans, Kids-girls-dress and Women-jeans as shown in Table 1. This well-trained model is able to identify products from well-curated datasets. In contrast, the OIDV4 dataset had much lower precision and recall (e. g. , 10. 2% for belts as shown in Table 2), pointing to issues with inaccurate bounding box annotations and varying image quality. On the custom-curated dataset, YOLOv9m achieved precision of 96. 6%, recall of 95. 5%, and a mean average precision of 97. 8%, showing its effectiveness in detecting e-commerce product images. YOLOv9 models generally outperformed YOLOv8, where as the YoloV9 model has improvements architecture like Path aggregation network (PANet), CSPDarkNet53, PGI, and SPPELAN.

As shown in the Table 3, YOLOv9m needed more computational resources (76. 5 GFLOPS) compared to YOLOv8m (69. 5 GFLOPS), which aligns with its better detection performance. Despite YOLO-NAS-M's competitiveness, YOLOv9m excelled, especially with the custom dataset, highlighting its efficiency. In comparison with the various thresholds such as mAP50 and mAP50-90 demonstrated the

various Yolo models robustness. Whereas the models trained on the OIDV4 dataset had lower mAP score values due to its inconsistent labelled images and bounding boxes in the dataset compared to the custom curated e-commerce dataset stressing the importance of the quality of the image data, annotations resulting in the model classification. As shown in the Figure 3, the model is showing good convergence with decreasing loss consistently and increased accuracy in the training and validation. Whereas in distribution focal loss distribution trend with some oscillations are shown but model is stabilized with tuning of the hyperparameters during the model training and overcomes the overfitting issue. Further as shown in the Figure 4, model trained with YoloV9-m shows the effective detections with higher confidence score for the various classes with right bounding boxes and accurately encapsulate the products in the image.

#### V. CONCLUSIONS

This work highlights the custom object detection, recognition, and classification using two e-commerce datasets, such as the OIDV4 apparel images and a custom-curated e-commerce image dataset, utilizing the YOLOv9, YOLOv8, and Yolo-NAS algorithms. The existing OIDV4 dataset, although varied, suffers from subpar bounding box quality, resulting in lower performance across the tested algorithms. Conversely, the custom e-commerce dataset, which features meticulously annotated bounding boxes and optimized hyperparameters for the YOLOv9m model, achieved notable results: 96.6% precision, 95.5% recall, and 97.8% mean average precision (mAP). This model was specifically designed to enhance performance and provide consistent recognition of e-commerce products by incorporating various image aspect ratios from the datasets. Compared to the YOLOv9m model, excelled the YoloV8 and Yolo-NAS in terms of computational efficiency and precision. Despite the promising outcomes, the current study mostly depends on the quality of the annotated datasets, which requires rigorous and precise labeling during the annotation process. Further, incorporating the semi-supervised or unsupervised learning techniques, and extending to a wide variety of product categories and different imaging scenarios would be beneficial to enhance the model's robustness and applicability in e-commerce contexts.

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